

Database Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity.

First Normal Form (1NF):

- Each column contains atomic values
- Each record is unique

Second Normal Form (2NF):

- Must be in 1NF
- No partial dependencies (non-key attributes depend on the whole primary key)

Third Normal Form (3NF):

- Must be in 2NF
- No transitive dependencies (non-key attributes depend only on the primary key)

Boyce-Codd Normal Form (BCNF):

- Stricter version of 3NF
- Every determinant is a candidate key

The reviewer noted that normalization is essential for maintaining data consistency in DBMS systems.